

Project Guide & Glossary

What this project does, how it does it, and what every number and term on the dashboard means.

SCOPE	Credit-risk pipeline (Python / DuckDB / scikit-learn) + CREDENCE dashboard (React)
DATASET	Lending Club accepted loans, 640,919 terminal loans, issued 2012–2016
MODEL	Calibrated gradient-boosted PD model, out-of-time tested, AUC 0.690
DATA POLICY	Every figure is labelled REAL (measured) or DEMO (synthetic, illustrative)

1. WHAT THIS PROJECT DOES

A credit-risk pipeline, and a dashboard built on top of it

This project has two parts. The first is a **credit-risk modelling pipeline** — Python, DuckDB, and scikit-learn — that takes a real portfolio of consumer loans, engineers a feature set, trains a probability-of-default (PD) model, calibrates it, backtests it against what actually happened to those loans, and checks it for fair-lending issues. The second is **CREDENCE**, a React dashboard that presents the pipeline's output the way a credit-risk or portfolio-management team would actually use it: portfolio health, borrower-level risk, an approval-threshold simulator, and model-monitoring.

The dataset is Lending Club's public “accepted loans” data — 640,919 36-month personal loans, issued between January 2012 and February 2016, each one already resolved (paid off or charged off) by the snapshot date. Every headline number in CREDENCE — the \$8.2B of exposure, the 14.1% default rate, the model's 0.690 AUC — is a real, reproducible measurement from this dataset and this pipeline, not a mock-up.

Two kinds of data, always labelled

Some things a real risk dashboard needs — a searchable table of individual loan applications, a day-by-day monitoring feed, per-borrower “why is this risky” explanations — don't exist at the row level in this public dataset. Rather than fabricate them silently, CREDENCE generates a seeded, labelled synthetic set for exactly those panels, and marks every one of them with a visible badge. Two tags recur through this document and the dashboard itself:

REAL

A measured result from the pipeline running against the actual Lending Club data — a portfolio aggregate, a model metric, a backtested outcome. Reproducible from the pipeline's own code.

DEMO

Seeded synthetic data generated to illustrate a capability the public dataset can't support at row level (individual applications, day-to-day monitoring). Never presented as a real result.

The full accounting of which panel is which lives on the dashboard's own **Data / Model** page, and is reproduced in section 4 of this guide.

Who this is for

- A credit-risk or portfolio-analytics team deciding where risk concentrates in a loan book.
- A credit-decisioning team weighing where to set an approval threshold and what that trade-off costs.
- A model-risk / validation function checking a PD model's discrimination, calibration, and stability.
- Anyone evaluating the underlying modelling work itself — the pipeline is built to the standard of an actual bank model-risk review: leakage-guarded features, an out-of-time test, a benchmark, a fairness check.

From raw loan data to a decision on the screen

The pipeline runs in five stages, each one a checkpoint a real model-risk review would ask for. The output of each stage is a file — DuckDB tables, then JSON artifacts — that the dashboard reads directly; nothing is invented at the presentation layer.

1

Load & stage

Lending Club's raw "accepted loans" CSV is loaded into DuckDB and staged: dates parsed, rates and percentages cast to numbers, loans restricted to 36-month term and a maturity window (2012–2016 originations) where at least 98% of that month's loans have already reached a final outcome — so the model is trained on loans whose fate is actually known, not still-open ones.

2

Engineer features, guard against leakage

About 33 fields knowable at underwriting time — FICO, income, DTI, credit-bureau attributes, loan purpose — go into an explicit allow-list. A second, explicit forbidden-list blocks anything that could leak the outcome or Lending Club's own risk grade into the model, and the pipeline asserts this at training time, not just by convention.

3

Split out-of-time, train, calibrate

Loans issued before 2015 are the training set; loans issued from 2015 onward are held out as the test set — the model never sees the future relative to what it's trained on, which is the realistic way a bank actually deploys a model. A gradient-boosted classifier is fit and then isotonic-calibrated, so its output can be read as an actual probability, not just a ranking score.

4

Benchmark, backtest, check fairness

The model is compared against a logistic-regression alternative and against Lending Club's own risk grade, with a statistical test (DeLong) on whether the improvement is real. Separately, every point on a 60-step approval-threshold curve is backtested against what actually happened to those loans — not just what the model predicted. And approval rates are checked across income, region, and home-ownership groups for disparate impact.

5

Publish artifacts, render the dashboard

Every result above is written to a JSON artifact. CREDENCE's data layer transcribes those artifacts verbatim into typed constants — there is no live re-computation in the browser, and no way for the dashboard to drift from what the pipeline actually measured.

Why an out-of-time split matters

Most of the numbers on the Monitoring page compare **train** (older, 2012–14 loans) against **test** (newer, 2015–16 loans). This is the single most important discipline in the whole pipeline: a model that only looks good on loans it was trained on tells you nothing about how it will perform on the next loan that walks through the door. Every headline metric in this guide — the 0.690 AUC, the calibration chart, the backtest

curve — is the **test**, out-of-time number, not the easier training-set number.

Where the model stops and the dashboard begins

CREDENCE adds nothing to the risk assessment itself. Its own logic is limited to: reading the pipeline's JSON artifacts, formatting and charting them, letting you move an approval-threshold slider along the pipeline's own backtested curve (interpolated, never extrapolated past what was actually measured), and generating the clearly-labelled synthetic borrower set and monitoring feed for the panels the public dataset can't support at row level.

3. THE DASHBOARD

A tour of the five pages

Overview — The whole book at a glance

Five headline numbers (exposure, default rate, high-risk accounts, expected loss, risk trend), the portfolio's risk-band distribution as a single proportional bar, a fair-lending flag if any approval group fails the 4/5ths rule, a risk-over-time chart, a breakdown of where expected loss concentrates (grade, purpose, credit score, vintage), and a monitoring alert feed.

Portfolio — Where exposure and risk sit across the book

The same portfolio sliced every way a risk committee would ask for: by loan purpose, by Lending Club grade, by credit-score band, by income band, by origination vintage, by region, and by loss-given-default per grade. Six filter dimensions highlight a chosen segment across every chart on the page.

Borrowers — Individual applications, drill-down detail

A sortable, searchable table of synthetic loan applications. Opening one shows a full financial and credit profile, a bar chart of what's pushing that borrower's risk score up or down, and the decision that resulted.

Decisions — What happens if the approval line moves

The centrepiece is a threshold simulator: drag a slider along the pipeline's real backtested curve and watch approval rate, exposure, expected loss, and realised profit move together — this is not a toy model, it's the actual out-of-time backtest. Below it, the real approval-rate gap by income band, and predicted-vs-realised default and loss by decision.

Monitoring — Is the model still working, and what does it lean on

Twelve-month trend tiles, an emerging-risk timeline, and the Model Risk section: discrimination (AUC/Gini/KS), calibration, train→test stability, a real global feature-importance ranking, and a benchmark against the lender's own risk grade with a significance test.

Two more pages

Data / Model is the accountability page: dataset and model provenance, a full REAL/DEMO ledger for every panel in the dashboard, and a stated list of what this build deliberately does not fake — authentication, live production monitoring, per-loan adverse-action reason codes, a true protected-class fair-lending test, and macro stress scenarios. **Settings** lets you move the display risk-band cut-offs (a labelling convenience) without touching the real underwriting cut-off used in Decisions.

4. GLOSSARY

Every term and number on the dashboard, explained

Organised by the kind of question each term answers. Where a term corresponds to a real, current figure in CREDENCE, that figure is given alongside the definition.

The core risk numbers

Exposure / EAD (exposure at default)	The dollar amount actually at risk on a loan — here, the amount funded at origination. Portfolio total: \$8.16B across 640,919 loans.
PD — probability of default	The model's estimated chance a given loan ends in default, expressed 0–100%. This is the number the whole platform is built around; every risk band, decision, and simulator reading is a function of PD.
LGD — loss given default	Of the money outstanding on a loan that defaults, the share the lender doesn't get back. Measured here from real recovery data (principal repaid plus recoveries, against funded amount) on charged-off loans — 49.95% portfolio-wide, ranging 44% (grade A) to 60% (grade G).
EL — expected loss	The portfolio's expected dollar loss: $PD \times EAD \times LGD$, summed across loans. Portfolio total: \$524.2M , 6.4% of exposure.
Default rate	The share of loans that actually defaulted (observed), as opposed to PD, which is the model's prediction before the fact. Portfolio observed rate: 14.09% .
Risk band (Low / Medium / High / Critical)	A loan's PD sorted into one of four buckets for reading at a glance: Low <5%, Medium 5–10%, High 10–20%, Critical 20%+. The cut-offs are configurable in Settings as a display convenience — moving them re-labels the bands but doesn't change any underlying number.

Judging the model

AUC (area under the ROC curve)	How well the model ranks defaulters above non-defaulters, from 0.5 (no better than a coin flip) to 1.0 (perfect separation). Out-of-time test result: 0.690 — meaningfully, but not enormously, better than the lender's own risk grade alone (0.669).
Gini coefficient	A rescaling of AUC onto a 0–1 discrimination scale: $Gini = 2 \times AUC - 1$. Test result: 0.381 .

KS statistic (Kolmogorov–Smirnov)	The largest gap between the cumulative share of good loans and the cumulative share of bad loans, as you sort by model score — a second, classic view of how well the model separates the two groups. Test result: 0.275 .
Brier score	The mean squared error between predicted probability and actual outcome (0 or 1) — lower is better. Rewards calibration, not just ranking. Test result: 0.119 .
Calibration	Whether a stated probability is actually true on average — if the model says 15% PD for a group of loans, about 15% of them should actually default. Shown as predicted-vs-observed default rate across ten equal-sized risk deciles; CREDENCE's model tracks within 1–2 points across most of the book.
Decile	The test loans sorted by predicted PD and split into ten equal-sized groups (worst decile to best), used to check calibration and lift group by group instead of on one blended average.
Precision / Recall / F1 (at a chosen PD flag)	At the PD $\geq$ 20% “high-risk” flag: Precision is the share of flagged loans that really did default (29%). Recall is the share of all actual defaults the flag caught (39%). F1 is their harmonic mean (0.33). All three move if the flag threshold moves — they describe one operating point, not the model overall.
Base rate	The share of loans in a given sample that actually defaulted — the denominator every lift and precision number is read against. Test-set base rate: 14.9% .
Out-of-time (OOT) split	Training the model only on older loans and testing it only on newer ones, mirroring how a model is actually used — scoring the future, not the past. Split here at loans issued 2015-01-01.
Train → test stability	How much a metric changes between the training-period result and the out-of-time test result. A model that holds up well shows only a small drop; CREDENCE's AUC softens from 0.715 to 0.690 — a modest, expected seasoning effect, not a red flag.
Benchmark	A simpler alternative the model is compared against, to prove the complexity is earning its keep. Here: a logistic-regression version of the same model, and Lending Club's own risk grade used as a naive PD estimate on its own.
DeLong test	A statistical significance test for whether one model's AUC is really higher than another's, or just higher by chance on this particular sample. CREDENCE's model beats risk-grade-alone with $z \approx 19.4$, $p \approx 2 \times 10^{-83}$ — not noise.

Feature importance (permutation)	How much a model's AUC drops when one input's values are randomly shuffled, holding everything else fixed — the bigger the drop, the more the model actually relies on that input. Computed here directly on the trained model, not assumed. Credit score, recent credit-seeking, and loan-to-income rank highest.
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The approval decision

Approval / underwriting cut-off	The PD threshold below which a loan would be approved. The pipeline's real, backtested cut-off is PD < 15% — distinct from the risk-band Critical threshold (20%), which is a monitoring classification, not an approval line.
Threshold simulator / backtest curve	A slider that moves along 60 real, measured points: at every possible cut-off, what approval rate, default rate, and profit actually resulted when applied to the out-of-time test loans' real outcomes. Nothing on this curve is a projection — every point already happened.
Realised vs. model (expected) loss/profit	“Model” figures are what the PD model predicted before the fact; “realised” figures are what actually happened to those loans. Realised credit loss runs about 13% above the model's expected loss across the curve — the model is directionally right but slightly optimistic on dollar terms.
Regret	The extra dollar cost of following the model's own profit-maximising cut-off instead of the cut-off that, with hindsight, actually maximised realised profit. Here: \$180,521 on an \$8B book — small relative to the portfolio, meaning the decision is sound even though the model isn't perfectly calibrated at the margin.
Loss ratio	Realised credit loss divided by model-expected loss, averaged across the backtest curve — a single number for “how optimistic is the model's dollar forecast.” Here: 1.126 , i.e. about 13% understated.

Fair lending & disparate impact

Adverse-impact ratio (AIR)	The lowest approval rate among groups on some dimension, divided by the highest — a standard regulatory screen for whether a policy disproportionately excludes one group. 1.00 is perfectly even.
4/5ths rule	A common enforcement guideline: an AIR below 0.80 is treated as a flag worth investigating. In this portfolio, income band (AIR 0.42) and home ownership (0.68) both fail; region (0.96) passes.
Disparate impact vs. disparate treatment	Disparate <i>impact</i> is a policy that is neutral on its face but produces an uneven outcome across groups — what the AIR check above measures. Disparate <i>treatment</i> would mean explicitly deciding differently based on a protected characteristic; nothing here tests for that directly.
Protected-class fair lending (ECOA)	A true fair-lending test under the Equal Credit Opportunity Act checks outcomes by race, sex, age, and other protected characteristics. Lending Club's public data has none of these fields, so this project can only run a proxy check on income, region, and home ownership — stated plainly as a limitation, not glossed over.
Adverse-action reason codes (Reg B)	The specific, individual reasons a lender must legally give an applicant who is declined. CREDENCE shows a real <i>global</i> feature-importance ranking (which inputs matter most, on average) but does not generate real <i>per-loan</i> reason codes — that would need a per-decision explainer (such as SHAP) run against every individual application, which is out of scope here.

Reading the loan book

Grade (A–G)	Lending Club's own letter risk grade at origination, used here only as a benchmark and for the loss figures — never as a model input, since it's built from the same information the model already sees (using it as a feature would be circular).
Vintage	The calendar year a loan was originated. Comparing vintages shows whether newer loans are performing better or worse than older ones — a seasoning or underwriting-drift check.
FICO / credit score band	The applicant's credit-bureau score at application, grouped into ranges. The single strongest input to the model by a wide margin.
DTI — debt-to-income ratio	Monthly debt payments divided by monthly income — a standard affordability measure and one of the model's stronger inputs.
Revolving utilisation	How much of a borrower's available revolving credit (credit cards, mainly) is currently in use. High utilisation is a classic distress signal.

Loan-to-income ratio	Loan amount divided by annual income — how large the loan is relative to what the borrower earns.
Home ownership	Whether the applicant rents, owns outright, or carries a mortgage — used both as a risk input and as one of the three fairness-check dimensions.
Loan purpose	What the borrower says the loan is for (debt consolidation, credit card, home improvement, etc.) — both a risk input and one of the sharpest portfolio-concentration lenses.

Watching the book over time

PSI — population stability index	A standard measure of how much the applicant population has drifted from some baseline — used to catch a model going stale because the world it's scoring has changed. Shown on Monitoring as an illustrative synthetic series (no live applicant stream exists here to measure it for real).
Delinquency rate	The share of loans currently behind on payments but not yet charged off — an early-warning signal that leads default rate.
Emerging-risk signal	An alert that some monitored metric crossed a defined threshold — illustrative on this build, generated the way a real monitoring pipeline's alerts would look, not from live data.

Reading the dashboard itself

REAL tag	A measured result from the pipeline, reproducible from its code and data.
DEMO DATA badge	Seeded synthetic data, generated because the public dataset has no row-level equivalent — always visibly marked, never presented as a measured result.
Risk spectrum	The proportional bar on Overview showing what share of funded exposure sits in each risk band — the single fastest answer to “how risky is the book right now.”
Contribution bars	On a borrower's detail view, a bar chart of which inputs are pushing that individual's risk score up (orange) or down (green). Built on the synthetic borrower set's transparent scoring function, not the pipeline's trained model — labelled DEMO for that reason.

5. WHAT THIS ISN'T

Stated limitations, not silent gaps

A few things this build deliberately does not simulate, because a fake version would be worse than stating the gap plainly:

- **Access control** — no authentication; anyone with the URL sees every panel.
- **Live model monitoring** — trend tiles are illustrative synthetic series, not scored production traffic (there is no production traffic).
- **Per-loan reason codes** — the feature-importance ranking is a real, global, model-level result; individual adverse-action notices would need a real per-decision explainer.
- **Protected-class fair lending** — the dataset carries no race, sex, age, or national-origin fields, so only a proxy disparate-impact check is possible.
- **Macro / stress scenarios** — the model reflects one historical window; there is no scenario engine for a rate shock or a recession.

CREDENCE is built over the public Lending Club dataset and this project's own reproducible pipeline. No claim is made about any real bank, customer, or portfolio.